SDG IMPACT HACKATHON 2026

# PROJECT AXIORA

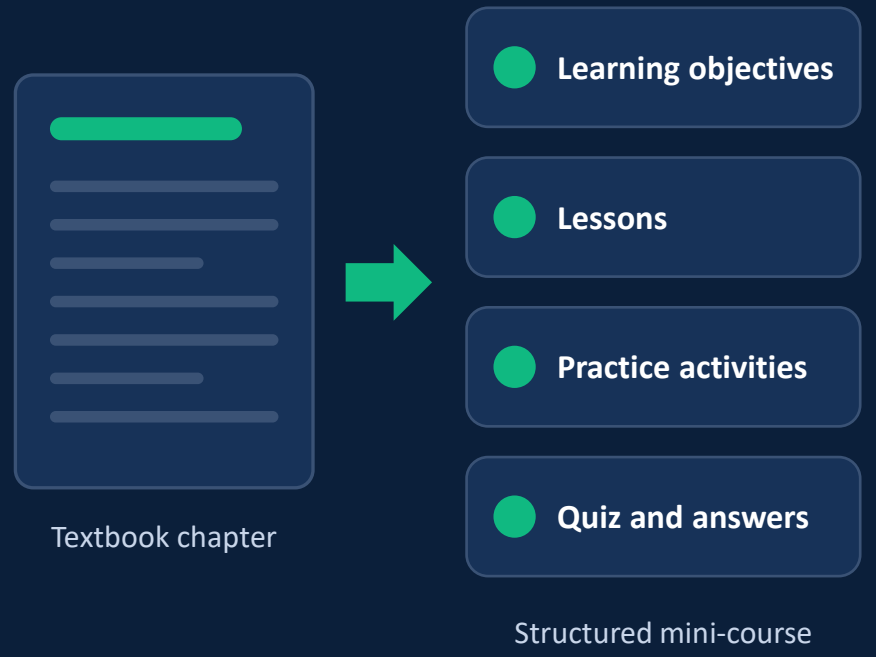
## CourseForge AI

From Textbooks to Structured Learning Experiences

**Team KayCode**

Presenter: Kayes Md Golam Murtuja

Al-Farabi Kazakh National University




**Track: AI for Education & Learning**

**SDG 4: Quality Education**

November 29, 2026 | Kazakh-British Technical University (KBTU), Almaty

PROBLEM STATEMENT

# Textbooks are rich in content, but not ready to teach

Teachers still convert chapters into lessons, activities and assessments by hand.

**EXISTING TEXTBOOK**

Chapters and pages of dense source material

**MANUAL PREPARATION**

Objectives, lesson order, examples, exercises, quizzes and source checks

**LESSONS, ACTIVITIES, ASSESSMENTS**

Organized teaching resources ready for the classroom

## Who faces this

- School teachers
- University instructors
- Tutors and independent educators
- Institutions digitizing resources

*Students benefit as secondary users.*

## Why it matters

- Time spent restructuring content is time not spent teaching
- Summaries shorten text but do not build a course
- Materials must stay faithful to the source

**THE OPPORTUNITY**

**Let AI draft the structure. Let teachers review, correct and approve it.**

PROPOSED SOLUTION

# CourseForge AI turns a chapter into a mini-course

A guided workflow from textbook PDF to a course the teacher reviews before use.

1. **Upload textbook PDF**
   Text extracted with page numbers
2. **Select chapter**
   Teacher picks the section to teach
3. **Generate mini-course**
   An LLM drafts a structured course
4. **Review and edit**
   Teacher corrects and approves

## What the teacher receives

- ✓ **Learning objectives** — Clear goals for each lesson
- ✓ **Structured lessons** — Explanations in a logical order
- ✓ **Examples and practice** — Worked examples and activities
- ✓ **Quizzes with explanations** — Answers with reasoning
- ✓ **Source-page references** — Links back to textbook pages
- ✓ **Editable before use** — Nothing is final until approved

## How it solves the problem

- Drafts the structure teachers now build by hand
- Keeps outputs traceable to source pages
- Leaves every final decision with the teacher

INNOVATION AND UNIQUENESS

# Not just a summary: a structured, reviewable course

*“Do not just summarize a textbook. Transform it into a structured, reviewable learning experience.”*

### Generic summarization

- Condenses document content
- Usually focuses on summaries
- May not provide a complete teaching structure

vs

### CourseForge AI (proposed)

- ✓ Generates structured lessons
- ✓ Defines learning objectives
- ✓ Creates practice activities and quizzes
- ✓ Links generated material to source pages
- ✓ Supports teacher review and editing

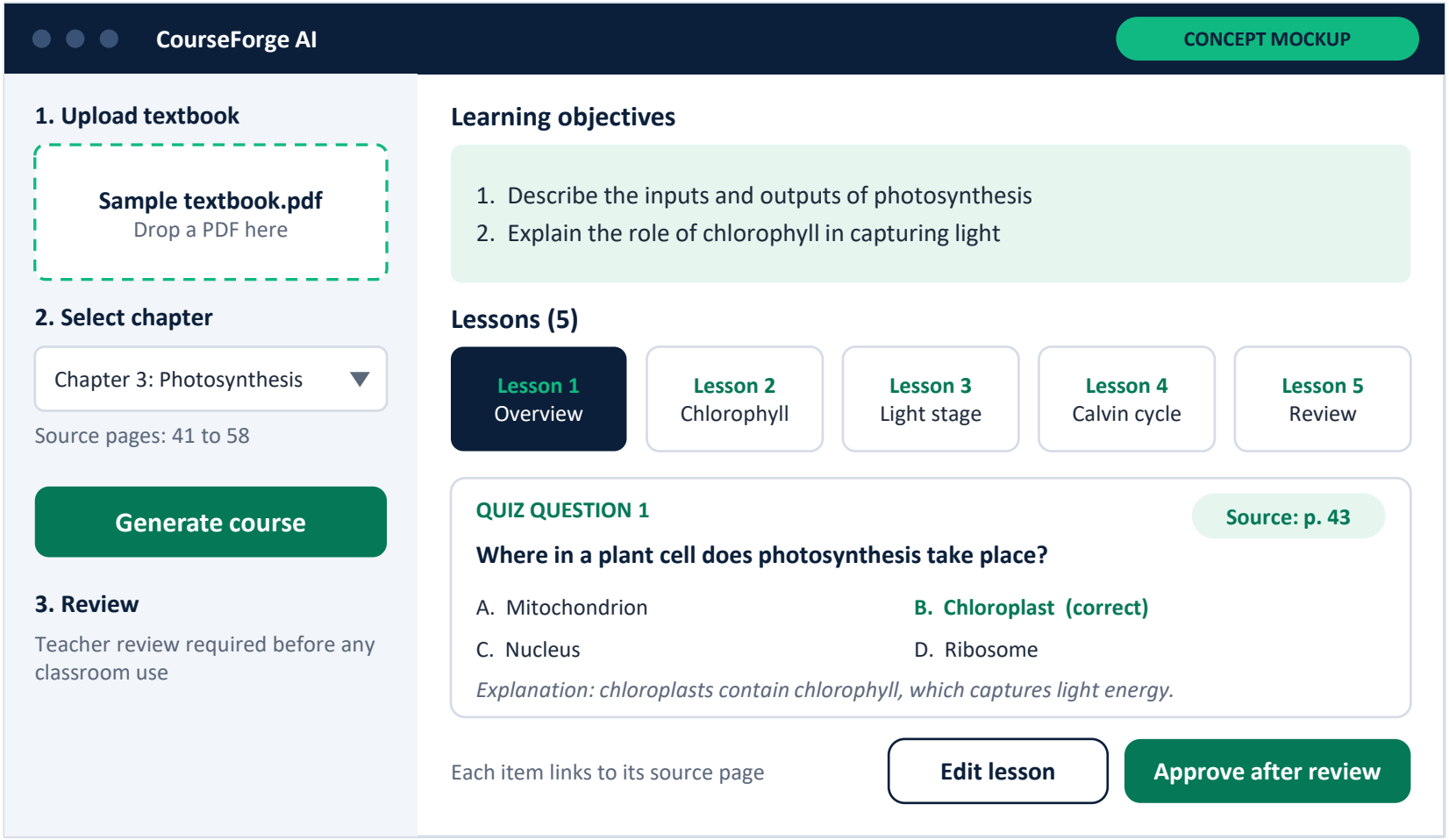
**Structured course generation** + **Traceability to source pages** + **Teacher control**

*Page references support verification but do not guarantee correctness. Teacher review remains essential.*

PROTOTYPE / MVP

# Proposed MVP Interface

Conceptual design with illustrative content. Not a screenshot of working software.




## Intended MVP scope

- ✓ One textbook chapter
- ✓ Five generated lessons
- ✓ Learning objectives
- ✓ Practice questions and quizzes
- ✓ Source-page references
- ✓ Teacher review before use

**Goal: demonstrate this workflow as working software at the hackathon.**

SDG ALIGNMENT AND IMPACT

# Supporting SDG 4: Quality Education

4

## QUALITY EDUCATION

*“Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.”*

Source: United Nations, sdgs.un.org/goals/goal4

## Intended benefits

### Less manual effort

Teachers start from a structured first draft

### Reuse existing textbooks

Chapters schools already have become digital lessons

### More accessible resources

Organized lessons and quizzes students can follow

## Planned evaluation metrics

Planned, not yet measured

1

### Teacher preparation time

Minutes to prepare one chapter, before and after using the tool

2

### Source-page reference accuracy

Share of generated references that point to the correct page

3

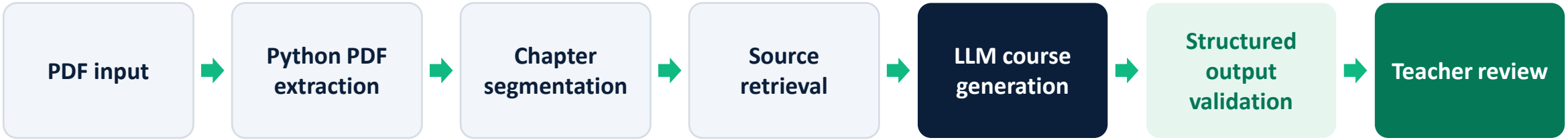
### Teacher acceptance

Share of generated materials accepted by teachers after review

FEASIBILITY AND IMPLEMENTATION

# A lean pipeline one developer can build

Proposed architecture with a deliberately limited first scope.




**Built with**

- Python
- Streamlit
- PDF extraction library
- LLM API
- Source-page metadata
- JSON output validation

## Deliberately limited scope

- One textbook chapter
- Five generated lessons
- Core workflow first
- No full LMS in version 1

*Small enough for an individual developer to prototype.*

## Key risks and mitigations

| | | |
|---|---|---|
| **Scanned or poorly formatted PDFs** | ➡ | Check extracted text quality first |
| **Inaccurate explanations** | ➡ | Mandatory teacher review before use |
| **Incorrect or missing citations** | ➡ | Validate references, show source pages |
| **Inconsistent output structure** | ➡ | Validate JSON against a fixed schema |
| **LLM API cost and response time** | ➡ | Small scope: one chapter, five lessons |
| **Copyright and permissions** | ➡ | Use permitted sample textbooks |

*Safeguards reduce errors but cannot eliminate them. Not yet deployed or tested.*

FUTURE DEVELOPMENT

# Roadmap: from one chapter to a teaching toolkit

1 — 2 — 3

PHASE 1

## Build and validate the MVP

One chapter, five lessons, quizzes, page references and teacher review.

PHASE 2 (FUTURE)

## Editing and export

Teacher editing, downloadable materials and support for more textbooks.

PHASE 3 (FUTURE)

## Scale and integrate

Multilingual support, curriculum alignment and LMS integration.

**“Turn the textbooks schools already have into learning experiences teachers can actually use.”**

Thank you | Team KayCode | Kayes Md Golam Murtuja | Al-Farabi Kazakh National University